Semantic Mapping of the 1981 and 1988 French Legislative Elections via Topic Modeling

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Abstract

This study presents a computational analysis of over 8,000 electoral manifestos from the French legislative elections of 1981 and 1988, archived in the ARCHELEC database maintained by Sciences Po. Given the impossibility of exhaustive manual reading at this scale, we apply Natural Language Processing (NLP) tools to extract dominant themes and map partisan positioning. Our pipeline combines **BERTopic**, built on contextual sentence embeddings (SBERT), with **UMAP** for two-dimensional projection, and selects the optimal number of topics via the C_v coherence score. The results reveal markedly different thematic structures across the two elections. In 1981, seven balanced topics capture a diversified political landscape encompassing labor, ecology, and programmatic themes. In 1988, a single dominant topic absorbs the vast majority of candidates, reflecting a dramatic compression of electoral vocabulary around a left-right opposition axis. We further observe the complete disappearance of ecology from the 1988 thematic structure, a partial convergence of governing party discourses, and the growing discursive marginalization of the Communist Party, confined to a niche topic centered on wage demands and legislative action.

1 Introduction

Electoral manifestos are a primary documentary source for the study of political discourse: written to address voters directly, they concentrate candidates' programmatic priorities and reflect the ideological cleavages of the moment. However, their manual analysis at scale is beyond the reach of individual researchers.

The French legislative elections of 1981 and 1988 represent two pivotal moments of the Fifth Republic. The 1981 election accompanied the presidential election of François Mitterrand and the historic left-wing alternation. The 1988 election marked the dissolution of Parliament and its renewal after the cohabitation period. Between these two elections, the political vocabulary evolved, new themes emerged (environment, employment, Europe) and certain parties strengthened or declined.

Our research question is the following: *How did the thematic priorities and ideological positioning of French political parties evolve between the 1981 and 1988 legislative elections, as reflected in candidates' electoral manifestos?*

To address this question, we built a three-step pipeline: extraction and normalization of manifestos from the ARCHELEC database, topic modeling via BERTopic, projection and visualization via UMAP. The code is available in a public GitHub repository[1].

[1] https://github.com/marieensae/ML_for_NLP_Final_Project

Preprint.

2 Data and Descriptive Statistics

2.1 Data Source

We use the ARCHELEC corpus, collected and maintained by Sciences Po as part of the project *Les archives électorales de la Ve République, du papier au numérique* [3]. This corpus assembles OCR-transcribed electoral manifestos from candidates in the legislative elections. Each document corresponds to one candidate's manifesto and is associated with metadata including the candidate's name, party affiliation, and constituency.

The texts are accessible from the SSP Cloud Minio repository as ZIP archives by year (`legislatives_1981.zip`, `legislatives_1988.zip`). The metadata are available in a centralized CSV file. After downloading and extraction, we match each text file to its metadata via the document URL.

2.2 Preprocessing and Cleaning

Several preprocessing operations are necessary before analysis. First, a non-negligible number of candidates have no party recorded in the metadata: these entries are excluded. Second, some candidates are affiliated with multiple parties (separated by `|` or `/`): we split these associations and retain one row per party. Third, we only keep parties with at least 30 manifestos for the visualization, in order to ensure statistically robust representation. Finally, we build an extended list of *stopwords* specific to this corpus, covering courtesy formulas, recurring proper nouns, and German terms present in Alsace-Moselle constituencies.

2.3 Descriptive Statistics

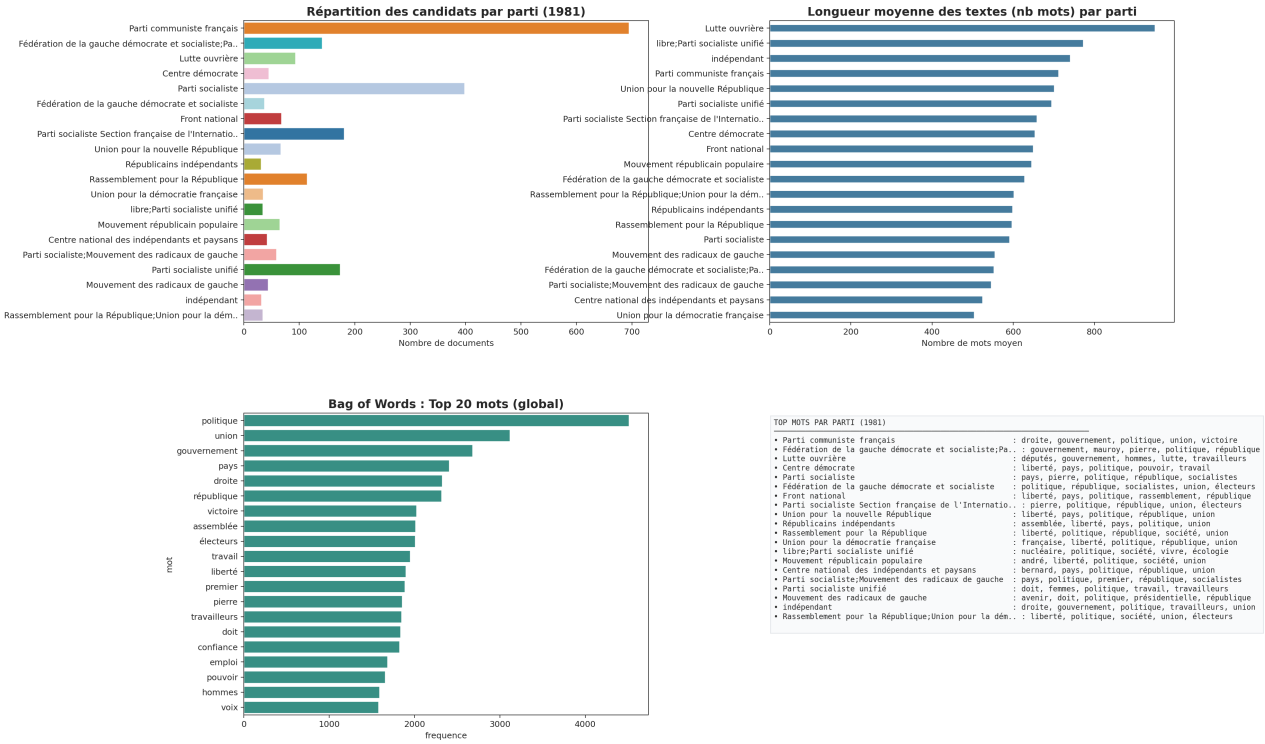

Figure 1: Descriptive statistics for the 1981 legislative elections. Clockwise from top left: distribution of candidates by party, average text length (number of words), top words per party, global Bag-of-Words (top 20 words).

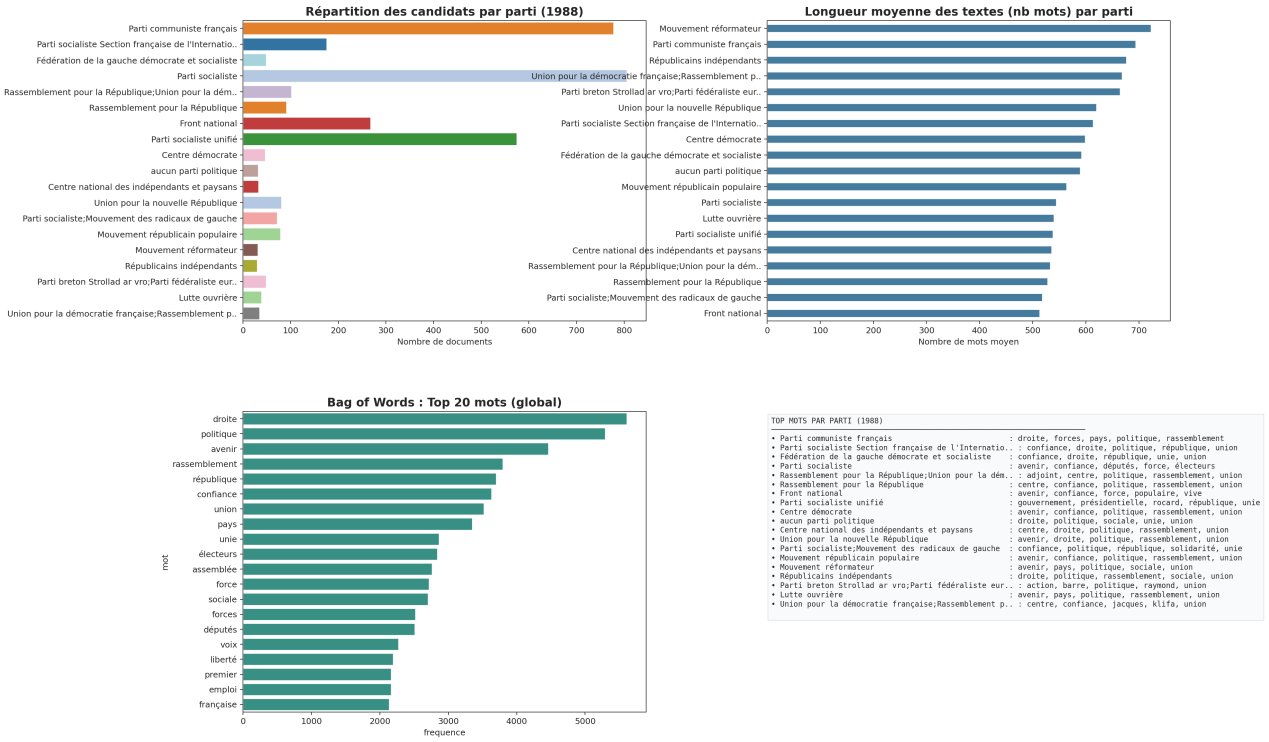

Figure 2: Descriptive statistics for the 1988 legislative elections.

Examining Figures 1 and 2 reveals several salient characteristics of the corpus. The French Communist Party (PCF) and the Socialist Party (PS) numerically dominate both elections in terms of candidates fielded. In 1981, the PCF fields the most candidates by a clear margin, while in 1988 the Parti socialiste unifié (PSU) surpasses all other parties. A marked fragmentation of the partisan landscape is visible, with many parties presenting only a few dozen candidates. Average text length varies more substantially across parties than previously: in 1981, Lutte ouvrière and independent candidates produce notably longer manifestos (over 800 words on average), whereas most established parties cluster around 600 words. In 1988, the Mouvement réformateur stands out with the longest average texts, followed closely by the PCF. The most frequent global words differ between the two elections: *politique* and *union* dominate in 1981, while *droite*, *politique*, *avenir* and *rassemblement* lead in 1988, reflecting the shift from a discourse of left-wing union to one of political competition and future-oriented mobilization. Top words per party confirm strong lexical specificities: the PCF mobilizes a vocabulary of *victoire* and class struggle in 1981, whereas in 1988 its discourse centers on *droite*, *forces*, and *pays*; the RPR consistently employs the semantic field of *force*, *rassemblement*, and republican values.

3 Related Work

3.1 Topic Modeling

Topic modeling refers to a family of unsupervised learning methods aimed at discovering latent thematic structures in a text corpus. The seminal method, **LDA** (*Latent Dirichlet Allocation*) [1], models each document as a mixture of topics, each topic being characterized by a probability distribution over the vocabulary. LDA has been widely applied to political discourse analysis [4]. Its main limitation is reliance on bag-of-words representations, which ignores word order and context.

NMF (*Non-negative Matrix Factorization*) [6] is a deterministic alternative to LDA that factorizes the TF-IDF matrix into two non-negative matrices. NMF tends to produce more parsimonious and interpretable topics than LDA, but suffers from the same limitations regarding contextual semantics.

BERTopic [5] represents a significant advance by combining pre-trained sentence embeddings (SBERT, [10]), a density-based clustering algorithm (HDBSCAN, [8]), and class-based TF-IDF for topic characterization. Unlike LDA and NMF, BERTopic captures the contextual meaning of terms, enabling it to distinguish thematically close but lexically distinct topics. Its application to political discourse analysis has proven particularly productive [2].

3.2 Dimensionality Reduction for Visualization

Projecting high-dimensional representations into a 2D space is a central problem for text corpus visualization. **t-SNE** [12] long remained the reference method, but suffers from several limitations: instability depending on perplexity parameters, poor preservation of global structure, and high computational cost.

UMAP (*Uniform Manifold Approximation and Projection*) [9] offers a faster and more stable alternative that better preserves the global structure of the point cloud while maintaining good resolution of local structures. UMAP has become the standard method for visualizing text embeddings [5]. The `n_neighbors` parameter controls the trade-off between local and global structure; we use `n_neighbors=15` with a cosine metric, which is particularly well-suited for textual similarity.

3.3 Application to Political Discourse Analysis

The application of NLP to electoral corpus analysis is a growing field. [4] provides a general framework for automated political text analysis. [11] applies vector-space models to measure ideological positioning from parliamentary debates. More recently, the advent of pre-trained language models has opened new avenues for mapping partisan positioning from electoral and parliamentary texts. Our contribution follows this line of work, applying these methods to a historically underexplored French corpus.

4 Methodology

4.1 General Pipeline

Our processing pipeline follows four successive steps:

1. **Extraction and formatting**: downloading archives, matching texts to metadata, filtering underrepresented parties, and building a corpus-specific *stopword* list.

2. **Semantic encoding**: transforming each manifesto into a dense 384-dimensional vector using the multilingual SBERT model `paraphrase-multilingual-MiniLM-L12-v2`.

3. **Topic modeling**: clustering documents in the embedding space via BERTopic (HDBSCAN) and characterizing topics via c-TF-IDF.

4. **Projection and visualization**: reducing to 2D via UMAP and mapping candidates colored by party affiliation and topic symbol.

4.2 Choice of Embedding Model

We use the `paraphrase-multilingual-MiniLM-L12-v2` model from Sentence-Transformers [10]. This choice is motivated by three reasons: (1) its multilingual nature handles the Alsatian manifestos partially written in German; (2) its compact size (12 layers, 384 dimensions) offers a good trade-off between semantic quality and computational cost on a corpus of several thousand documents; (3) its performance on French semantic similarity tasks has been validated in the literature [10].

4.3 BERTopic Configuration and Topic Number Selection

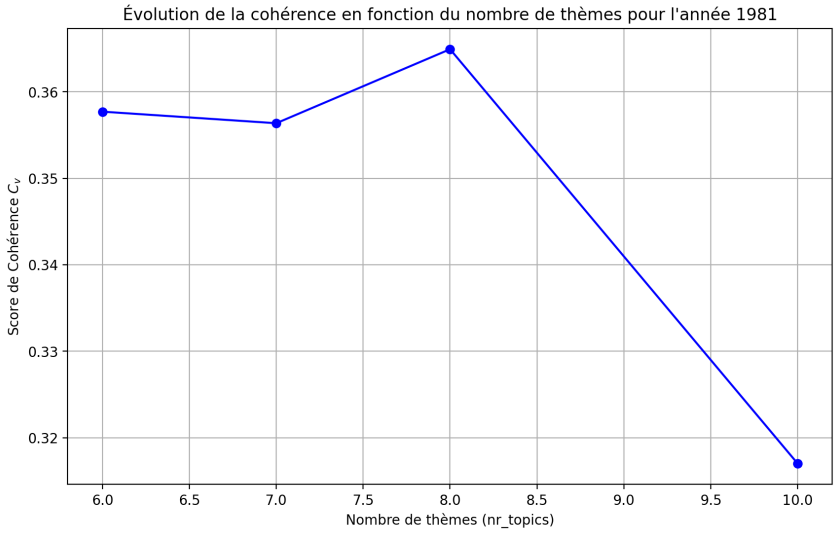

Figure 3: Coherence score (C_v) as a function of the number of topics (`nr_topics`) for the 1981 election. The curve peaks at 8 topics, indicating that this granularity maximizes intra-topic semantic cohesion.

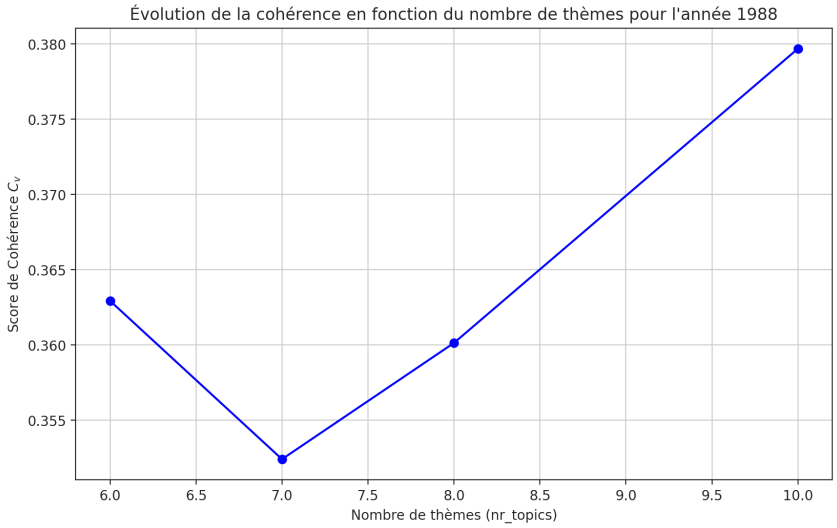

Figure 4: Coherence score (C_v) as a function of the number of topics (`nr_topics`) for the 1988 election. The curve exhibits a local minimum at 7 topics and then increases monotonically up to 10, suggesting that the thematic structure of the 1988 corpus supports finer granularity.

The choice of the number of topics is guided by the C_v coherence score, computed for values of `nr_topics` ranging from 6 to 10 (Figures 3 and 4). For 1981, the coherence curve peaks clearly at 8 topics ($C_v \approx 0.363$) before declining sharply, confirming this as the optimal granularity. For 1988, the curve presents a local minimum at 7 topics ($C_v \approx 0.353$) and rises monotonically thereafter, reaching its maximum at 10 topics ($C_v \approx 0.380$). We retain 9 topics for 1988 as a balance between maximizing coherence and preserving interpretability: a finer granularity beyond 9 risks producing topics that are too narrowly localized to carry substantive political meaning. After topic extraction by BERTopic, topics are reduced to the target number via hierarchical merging.

The `CountVectorizer` uses our extended stopword list for topic characterization. Documents unassigned to any main topic (topic -1 in BERTopic, corresponding to atypical points not grouped by HDBSCAN) are retained in the visualization so as not to bias the spatial representation.

4.4 UMAP Configuration

UMAP is applied directly to the SBERT embeddings (384 dimensions) with the following parameters: `n_neighbors=15`, `min_dist=0.1`, `metric='cosine'`, `random_state=42`. The cosine metric is preferred over Euclidean distance because SBERT embeddings are normalized vectors for which cosine similarity is more informative.

4.5 Candidate Selection for Display

To ensure map readability, we select at most 20 candidates per party for label display. For each party, we retain the candidates whose topic membership score is highest, which promotes the visual emergence of coherent partisan clusters. The full set of points (candidates) remains displayed as a scatter cloud.

5 Results

5.1 1981 Legislative Elections

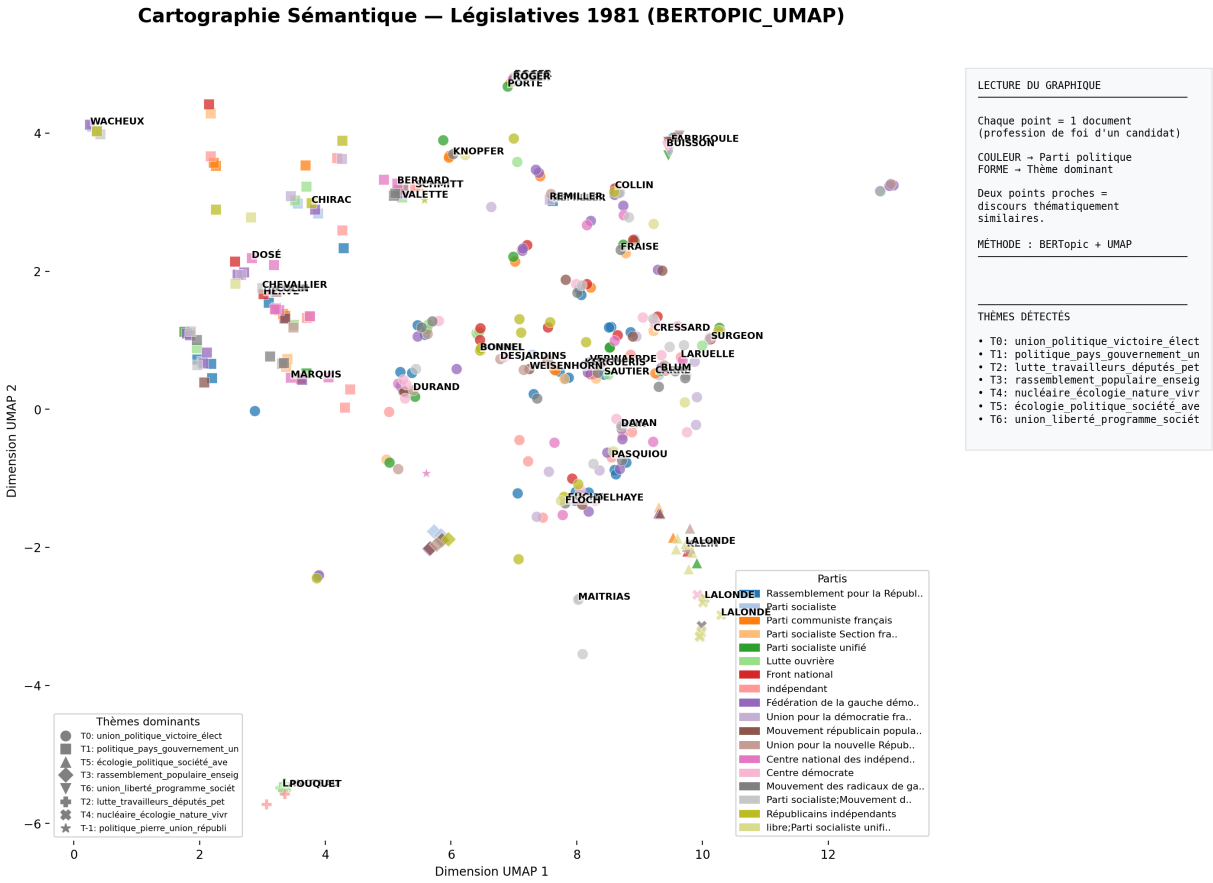

Figure 5: Semantic map of electoral manifestos, 1981 Legislative Elections. Each point represents one candidate. Color indicates the party; marker shape indicates the dominant topic identified by BERTopic. Points that are close together share thematically similar discourse.

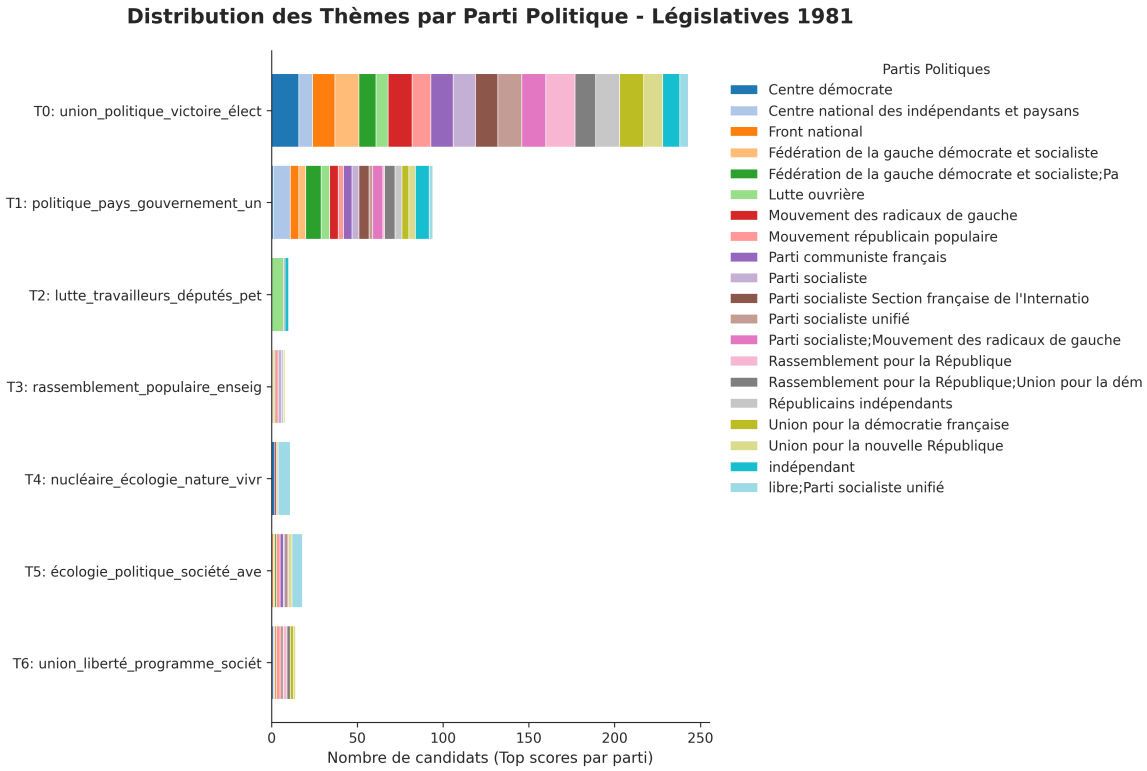

Figure 6: Distribution of dominant topics by party, 1981 Legislative Elections. Each bar represents the number of candidates (top topic score per party) assigned to a given topic.

BERTopic identifies seven main topics for the 1981 election (Figures 5 and 6). The topics can be interpreted as follows from their keywords:

- **T0 : Union, politics, and electoral victory** (*union, politique, victoire, élect*): by far the dominant theme, shared across virtually all parties. It reflects the overarching electoral mobilization context of 1981, characterized by the prospect of a left-wing alternation and the centrality of union-based discourse. The distribution chart confirms that T0 is nearly universal, accounting for the bulk of candidates from the PCF, PS, RPR, and most minor parties.
- **T1 : Government and country** (*politique, pays, gouvernement, un*): a second cross-cutting theme centered on governance and national priorities, with significant presence across left and right parties. It reflects a more programmatic discourse oriented toward policy content rather than electoral mobilization.
- **T2 : Workers and labor demands** (*lutte, travailleurs, députés, pet*): a minority theme concentrated among radical left candidates, particularly Lutte ouvrière, evoking class struggle and workers' representation.
- **T3 : Popular gathering and education** (*rassemblement, populaire, enseig*): a theme associated with centrist and Gaullist traditions of popular gathering, with some presence of educational themes. It is shared across several right-wing parties.
- **T4 : Nuclear energy and ecology** (*nucléaire, écologie, nature, vivr*): a niche theme reflecting the emerging ecologist movement, grouping candidates from ecologist parties alongside a few candidates from the liberal-ecological formation *libre;Parti socialiste unifié*.
- **T5 : Ecology, society, and future** (*écologie, politique, société, ave*): a broader ecologist-societal theme, partially overlapping with T4 but extending toward social and programmatic concerns. The distribution shows it captures candidates from a wider range of parties than the strictly anti-nuclear T4.

- **T6 : Freedom, programme, and society** (*union, liberté, programme, sociét*): a liberal-centrist theme emphasizing individual freedom and programmatic clarity. It is particularly present among Républicains indépendants and centrist formations.

The semantic map (Figure 5) reveals a broadly dispersed spatial organization. Unlike the 1988 map, the point cloud spans the full UMAP space without a single dominant cluster: this reflects the ideological fragmentation of the 1981 election, in which many parties presented competing but overlapping discourses. The right side of the map groups candidates with more ecologist and liberal-programmatic vocabularies (LALONDE and ecologist candidates appear at coordinates $(10, -2)$ and below), while the upper portion hosts candidates with more distinctive discourses (ROGER, PORTE at the top). The left part of the cloud is dominated by square markers (T1 — government/country theme), associated primarily with right-wing parties. The spatial isolation of Lutte ouvrière candidates (T2 — workers theme) confirms the lexical distance between radical labor discourse and the mainstream political vocabulary.

5.2 1988 Legislative Elections

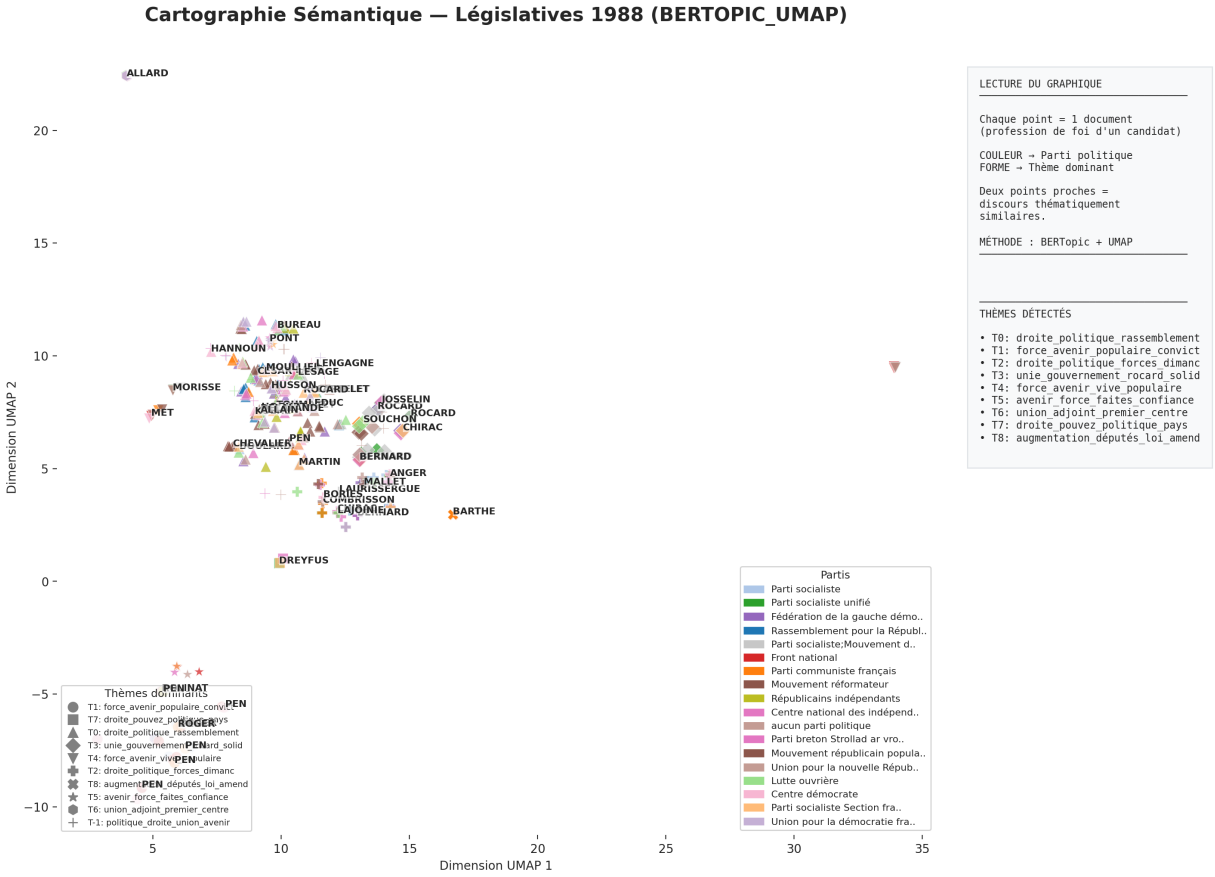

Figure 7: Semantic map of electoral manifestos, 1988 Legislative Elections. Each point represents one candidate. Color indicates the party; marker shape indicates the dominant topic identified by BERTopic.

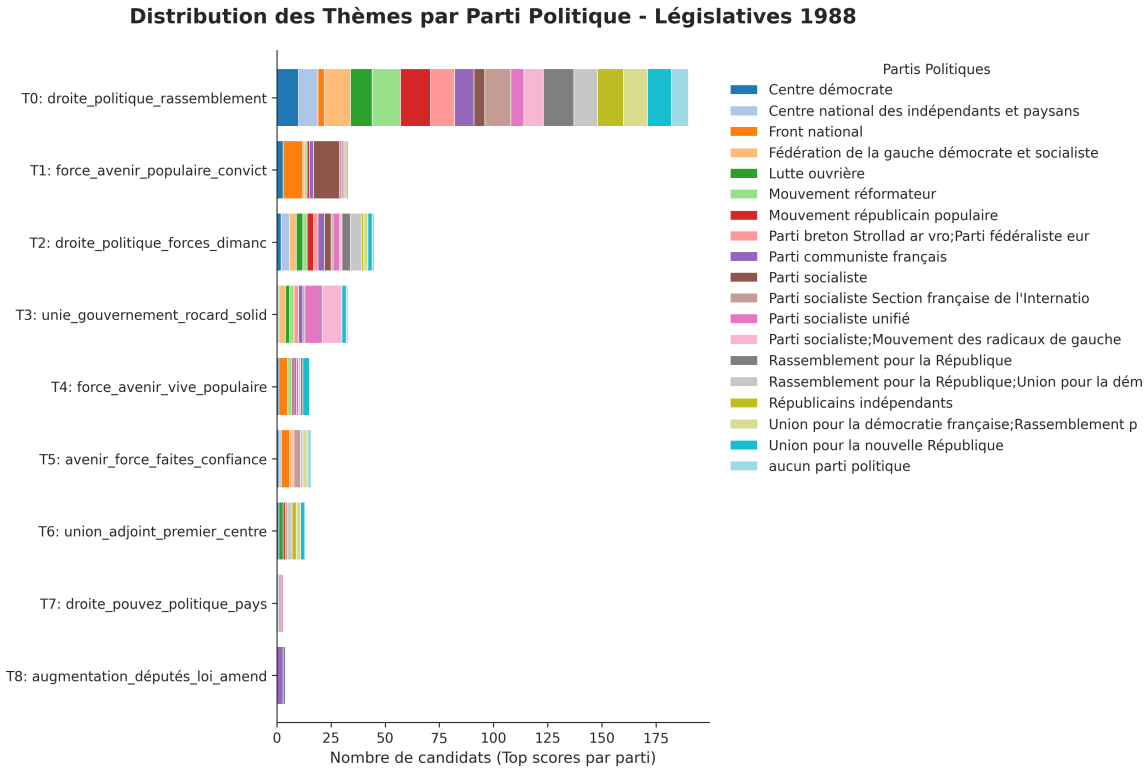

Figure 8: Distribution of dominant topics by party, 1988 Legislative Elections. T0 (*droite_politique_rassemblement*) is the overwhelmingly dominant topic, accounting for the vast majority of candidates across all parties.

For 1988, BERTopic identifies nine topics whose structure differs noticeably from that of 1981 (Figures 7 and 8):

- **T0 : Right, politics, and gathering** (*droite, politique, rassemblement*): the overwhelmingly dominant theme, capturing around 190 candidates and spanning virtually all parties. The distribution chart shows that T0 is so prevalent as to render all other topics marginal in comparison. Its keywords suggest that the 1988 electoral discourse is structured around a left-right opposition, with *rassemblement* as the shared mobilizing vocabulary.
- **T1 : Force, future, and popular conviction** (*force, avenir, populaire, convict*): a forward-looking mobilization theme, found primarily among Front national, Mouvement républicain populaire, and some Parti socialiste Section française de l'Internationale candidates. Its emphasis on *force* and *convict* suggests a more assertive, confrontational discourse register.
- **T2 : Right, politics, and forces (Sunday)** (*droite, politique, forces, dimanc*): a variant of T0, slightly differentiated by the presence of *dimanc* (Sunday), possibly reflecting religious conservative themes or reference to an electoral event. Shared across right-wing parties.
- **T3 : Unity, Rocard government, and solidarity** (*unie, gouvernement, rocard, solid*): a specifically socialist theme anchored in the governmental discourse of Michel Rocard's expected cabinet. The distribution confirms that this topic is concentrated among PS and allied candidates (Mouvement des radicaux de gauche), reflecting the governing majority's vocabulary of national unity and openness.
- **T4 : Force, future, and vivid popular appeal** (*force, avenir, vive, populaire*): another variant of right-wing mobilization discourse, with an emphasis on vivid popular energy (*vive*). Present among RPR-UDF formations.
- **T5 : Future, force, and voter confidence** (*avenir, force, faites, confiance*): a direct appeal to voter confidence with a forward-looking lexicon, shared between left-leaning and centrist

candidates. The imperative *faites confiance* is characteristic of an incumbent or moderate discourse seeking legitimization.

- **T6 : Union, deputy, and local centre** (*union, adjoint, premier, centre*): a strongly localized theme, associated with notable local candidacies featuring the vocabulary of deputies, local office holders (*adjoint*), and centrist networks.

- **T7 : Right, you can, politics, country** (*droite, pouvez, politique, pays*): a direct address discourse (*pouvez* = you can) deployed by right-wing candidates. Its small size reflects a specialized register of direct voter engagement.

- **T8 : Wage increases, deputies, and legislation** (*augmentation, députés, loi, amend*): a specifically PCF-associated theme, exclusively capturing Communist candidates with a focused vocabulary of wage demands, parliamentary representation, and legislative action. It appears only marginally in the distribution, reflecting the PCF's declining electoral footprint in 1988.

The 1988 semantic map (Figure 7) displays a more concentrated organization than that of 1981. The main cloud groups the majority of candidates in a central region (UMAP coordinates approximately 5–20 on dimension 1, 5–15 on dimension 2), with a partial overlap of governing party and opposition discourses which is a sign of convergence toward a shared institutional vocabulary. The PCF, whose topic T8 is associated with wage-and-legislation discourse, is located in a lower-left cluster, clearly separated from the mainstream cloud, reflecting its growing political isolation after the breakup of the left-wing union. The outlier *ALLARD* at coordinates approximately $(5, 22)$ corresponds to an atypical manifesto whose content strongly diverges from the rest of the corpus. Several candidates labeled PEN (Front national) appear in a lower subcluster around $(7, -8)$, confirming the Front national's spatial distinctiveness in the semantic map.

5.3 Cross-Election Comparison

Several notable evolutions emerge from comparing the two maps and topic distributions:

Thematic concentration. In 1981, no single topic accounts for more than approximately 250 candidates, and the distribution across topics is relatively balanced (Figure 6). In 1988, T0 alone absorbs nearly all candidates (approximately 190 out of a smaller filtered corpus), with all remaining topics accounting for at most 50 candidates each (Figure 8). This reflects a dramatic compression of thematic diversity: the 1988 campaign vocabulary converged around a dominant right-left-rassemblement axis.

Thematic recomposition. In 1981, the thematic field is structurally diversified, with specific topics for ecology (T4, T5), labor (T2), and freedom/programme (T6) alongside the dominant electoral themes. In 1988, ecology has entirely disappeared from the topic structure, replaced by variants of a right-wing mobilization theme, and the PCF's discourse is confined to a marginal topic (T8) focused on wage increases and legislation. This reflects the left's transition from an opposition posture with broad programmatic ambitions to one of accountability and track record.

Convergence of moderate discourses. The governing parties (PS, RPR, UDF) show greater spatial overlap in 1988 than in 1981, suggesting a partial homogenization of political vocabulary between the left and right of government which is what some analysts have termed convergent social-liberalism.

Marginalization of the PCF. The growing spatial separation of the PCF between 1981 and 1988 reflects its discursive isolation. In 1981, T2 (workers theme) remained a visible, if minority, component of the thematic landscape. In 1988, the PCF is confined to the marginal T8 and T7, with its vocabulary centered on specific demands (wages, legislation) that set it apart from the rest of the political spectrum. This is a consequence of the breakup of the left-wing union and the rise of the PS as the dominant party of the French left.

Persistence of outliers. Both maps feature isolated points corresponding to atypical manifestos: very short texts (fewer than 50 words after cleaning), texts in a foreign language, or standardized manifestos lacking programmatic content. These unfiltered cases are useful for identifying limitations in the quality of OCR in the corpus.

6 Discussion

6.1 Validity of the Approach

The BERTopic + UMAP combination produces maps whose structure is consistent with the historical knowledge of the political landscape of both elections. The main political families are correctly separated in the projection space, and the identified topics correspond to documented discursive realities. The coherence scores, while modest in absolute terms (ranging from 0.35 to 0.38), are consistent with values reported in the literature for short political texts with noisy OCR content. This constitutes a satisfactory qualitative validation in the absence of ground truth.

However, several limitations must be acknowledged. First, the variable quality of OCR introduces noise into the documents: some texts contain artefacts (isolated digits, special characters) that bias the embeddings. Second, UMAP is a stochastic method whose projections are not strictly comparable across years : two distinct maps cannot be directly superimposed to measure positional shifts. A joint projection (training UMAP on the concatenation of both corpora) would be necessary for a rigorous comparison of positions. Third, the strong dominance of T0 in 1988 may partially reflect a limitation of the hierarchical merging procedure: merging down to 9 topics from a larger initial topic set may produce one very large residual topic absorbing heterogeneous discourse.

6.2 Alternative Methodological Choices

We evaluated NMF and LDA as alternatives to BERTopic. Topics obtained by NMF are more easily interpretable in terms of keywords but less semantically coherent: documents addressing the same subject with different vocabulary end up in distinct topics. LDA produces results similar to NMF on this corpus, with increased instability depending on the number of topics chosen. BERTopic, thanks to its contextual embeddings, handles synonymy and polysemy more effectively, which is particularly valuable for political texts where the same concept may be expressed in multiple ways across parties.

For projection, t-SNE produces more compact visualizations than UMAP but with poorer preservation of inter-cluster distances, which hinders the interpretation of relationships between political families. UMAP is therefore preferred for its better global readability.

6.3 Future Directions

Several natural extensions of this work can be envisaged. Extending the analysis to other elections (1973, 1978, 1993, 2002) would allow long-term discursive trajectories to be examined. Using large language models such as CamemBERT [7] for encoding could improve the quality of embeddings for French text. A joint UMAP projection trained on the concatenation of both corpora would enable direct positional comparison across elections. Finally, a sentence-level rather than document-level analysis would allow finer-grained topic identification, potentially resolving the dominance issue of T0 in 1988.

7 Conclusion

This study demonstrates the feasibility and relevance of a computational approach to large-scale electoral discourse analysis. The BERTopic + UMAP pipeline applied to the manifestos of the 1981 and 1988 legislative elections reveals thematic structures consistent with the political history of the period, and identifies significant discursive evolutions between the two elections.

The main findings are: a dramatic thematic concentration in 1988 compared to 1981, where a single dominant topic absorbs the majority of candidates, reflecting the compression of political vocabulary around a left-right opposition axis; a thematic recomposition between 1981 and 1988, from a diversified register including ecology, labor, and freedom themes to a more homogeneous institutional register; a partial convergence of governing party discourses in 1988; a growing discursive marginalization of the PCF, visible in its spatial isolation and its confinement to a niche topic centered on wage demands and legislative action.

These results open avenues for the systematic study of French political discourse evolution across all elections of the Fifth Republic, a task made accessible by the richness of the ARCHELEC corpus and the power of contemporary NLP tools.

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